

The Two-Player $3n \pm 1$ Game: A Binary Normal Form and a Well-Founded No-Draw Proof

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<https://github.com/Grisha-Pochuev/3n-plus-minus-1-game>

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Abstract

Consider the impartial two-player game on positive odd integers in which a move from $n > 1$ chooses $3n - 1$ or $3n + 1$ and removes all powers of two; reaching 1 wins immediately. Arbitrary play need not terminate: for example $5 \rightarrow 7 \rightarrow 5$ is a legal cycle. The question is whether optimal play nevertheless has finite remoteness from every starting position. We conjugate the game to a binary normal form with an expanding move A and a strictly contracting move B , introduce constant-tail coordinates and a three-free source map, and rank marked proof configurations by retained source, finite proof-height tokens, and a typed fixed-fibre normalizer. The key entry step is a factor-removal lemma that either removes one factor of three at the same tail exponent or produces finite outcome provenance. This prevents an unranked return from a factor-free three/two frame to the preceding exponent-one state. A one-shot initialization then allows the typed normalizer to be entered without treating a larger temporary inner source as a source decrease. The resulting well-founded marked normalization excludes all draw positions.

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1 The game and the theorem

The current state is a positive odd integer n . For $n > 1$ the player to move chooses one of

$$3n - 1, \quad 3n + 1,$$

and divides the chosen value by the largest possible power of 2, leaving an odd integer. If the resulting odd integer is 1, the player who made the move wins immediately. Players alternate with perfect information.

A position is called WIN for the player to move if at least one move leads to LOSS. It is called LOSS if every legal move leads to WIN. These two classes are the inductively generated finite-remoteness classes. Any position not in either class is called DRAW.

Theorem 1.1 (Main theorem). *Every positive odd starting position has finite remoteness. Equivalently, the $3n \pm 1$ game has no DRAW positions, and optimal play reaches 1 after finitely many moves.*

The theorem is not a statement about arbitrary play. There are legal cycles, such as

$$5 \longrightarrow 7 \longrightarrow 5,$$

under suitable choices by the players.

2 Binary conjugation

Write an odd state as $n = 2m + 1$ and define

$$F(m) = \left\lceil \frac{3m}{2} \right\rceil.$$

For a nonnegative integer x , let $R(x)$ be the integer obtained by deleting the maximal alternating suffix of the binary expansion of x . Thus $R(1001_2) = 10_2$ and $R(10101_2) = 0$.

Proposition 2.1 (First normal form). *For $m > 0$, the two children in m -coordinates are exactly*

$$F(m), \quad F(R(m)).$$

Proof. One has

$$3(2m + 1) - 1 = 2(3m + 1), \quad 3(2m + 1) + 1 = 2(3m + 2).$$

Exactly one of $3m + 1$ and $3m + 2$ is odd. In the other expression, each additional factor of two removed corresponds to crossing one additional alternating bit at the end of the binary expansion of m . The process stops at the first repeated adjacent bit, which is precisely the deletion defining $R(m)$. Returning to m -coordinates gives the displayed pair. $\square$

Since both children lie in the image of F , we represent the state $F(q)$ by q . The conjugated game has terminal state 0 and moves

$$A(q) = \left\lceil \frac{3q}{2} \right\rceil, \quad B(q) = R(A(q)).$$

Proposition 2.2 (Strict contraction). *For every $q > 0$,*

$$B(q) < q.$$

Proof. Deleting a nonempty binary suffix gives

$$B(q) \leq \left\lfloor \frac{A(q)}{2} \right\rfloor \leq \left\lfloor \frac{3q + 1}{4} \right\rfloor < q$$

for $q \geq 2$, while $B(1) = 0$. $\square$

Thus all unbounded difficulty comes from the expanding branch A and from the unbounded alternating suffix removed by B .

3 Constant-tail coordinates and coefficient sources

For a positive odd integer a , an exponent $r \geq 1$, and a phase $e \in \{0, 1\}$, put

$$Q_r^e(a) = a2^r - e.$$

This is a binary word whose final constant block has length r and consists of the bit e .

Lemma 3.1 (Long-tail recurrence). *For every positive odd a , phase e , and $r \geq 3$,*

$$A(Q_r^e(a)) = Q_{r-1}^e(3a), \quad B(Q_r^e(a)) = Q_{r-2}^e(3a).$$

Moreover

$$B(Q_1^e(a)) = B(Q_2^e(a)).$$

Proof. For $r \geq 3$, the expanding child still ends in at least two equal bits; hence its maximal alternating suffix consists only of the final bit. This gives both long-tail identities. At the boundary, the binary words of $3a - e$ and $6a - e$ differ by appending one bit that extends the same alternating suffix, so the same prefix remains after deletion. $\square$

Define the embedded three-free coefficient

$$J(s) = 2A(s) + 1.$$

Every positive odd coefficient a has a unique factorization

$$a = 3^k J(s), \quad k \geq 0.$$

The integer s is called the coefficient source of a and is denoted $\text{src}(a)$.

Lemma 3.2 (Source boundary). *For every $s > 0$ there is a phase*

$$\alpha(s) = 1 - \left(\left\lfloor \frac{s}{2} \right\rfloor \bmod 2 \right)$$

such that

$$\text{odd}(3J(s) + 1 - 2\alpha(s)) = J(A(s))$$

with 2-adic valuation one, while the opposite phase selects $J(B(s))$ with valuation at least two.

Multiplying a constant-tail coefficient by 3 preserves its coefficient source. Passing through a factor-free signed boundary with coefficient $J(s)$ therefore exposes exactly one ordinary A/B move on the source.

4 Finite proof tokens

The terminal state 0 is LOSS with height $h(0) = 0$. A finite WIN state has height

$$h(x) = 1 + \min\{h(y) : y \text{ is a LOSS child of } x\},$$

and a finite LOSS state has height

$$h(x) = 1 + \max\{h(y) : y \text{ is a child of } x\}.$$

Only finite states whose outcome and height comparison have actually been established are inserted into the marked configuration. They are called proof tokens.

Lemma 4.1 (Multiset token rank). *For a finite multiset M of proof heights define*

$$\Phi(M) = \#_{h \in M} \omega^h,$$

the natural ordinal sum of the monomials ω^h . Replacing one token of height H by any finite multiset of certified descendants of heights strictly below H strictly decreases Φ .

Proof. In Cantor normal form the coefficient of ω^H decreases by one, and all inserted terms have lower exponent. No finite collection of lower terms can compensate for the loss at exponent H . $\square$

5 Factor removal at fixed tail exponent

The next lemma is the entry device that prevents a factorful exponent-one state from creating an unranked restart.

Lemma 5.1 (Fixed-tail factor removal). *Let c be positive and odd, $e \in \{0, 1\}$, and $r \geq 2$. Put*

$$X = Q_r^e(3c), \quad G = Q_r^e(c), \quad H = Q_{r+1}^e(c).$$

If X is DRAW, then either G is DRAW, or the local configuration supplies finite outcome provenance suitable for a marked typed entry. More explicitly, if $Y = A(G)$, then

$$\text{moves}(H) = \{X, Y\},$$

and, whenever G is not DRAW, at least one finite WIN/LOSS token is forced among G, Y and the other child of G .

Proof. Since $r + 1 \geq 3$, Lemma 3.1 gives

$$A(H) = Q_r^e(3c) = X, \quad B(H) = Q_{r-1}^e(3c).$$

For $r \geq 3$ this second state is $A(G)$ by the same long-tail recurrence. For $r = 2$, $G = Q_2^e(c)$ and $A(G) = Q_1^e(3c)$, again equal to $B(H)$. Thus $\text{moves}(H) = \{X, Y\}$.

Because X is DRAW, the state H cannot be LOSS. If H is WIN, then its other child Y is LOSS, giving a finite token. Assume therefore that H is DRAW. Then Y cannot be LOSS. If G is DRAW, we are in the first alternative.

It remains to suppose that G is finite. If G is LOSS, then all its children, including Y , are WIN. If G is WIN, then either Y is a LOSS witness or, since Y is not LOSS, the other child of G is a LOSS witness. In all cases the displayed local diamond provides an actual finite outcome token with proved incidence. $\square$

Corollary 5.2 (Removing all factor powers at exponent two). *Let $a = J(s)$ and $k \geq 1$. If*

$$Q_2^e(3^k a)$$

is DRAW, then either a finite provenance token is obtained, or

$$Q_2^e(a)$$

is DRAW.

Proof. Apply Lemma 5.1 repeatedly with $r = 2$ and $c = 3^{i-1}a$. If the token alternative never occurs, the factor exponent is reduced by one at each step until it reaches zero. $\square$

Lemma 5.3 (Exponent-one lift to exponent two). *Let a be positive and odd and $k \geq 1$. If*

$$R_k = Q_1^e(3^k a)$$

is DRAW, then either

$$P_{k-1} = Q_2^e(3^{k-1} a)$$

is DRAW, or P_{k-1} is WIN and its other child is a finite LOSS token.

Proof. The long-tail boundary identities give

$$A(P_{k-1}) = R_k.$$

Since P_{k-1} has a DRAW child, it cannot be LOSS. If it is not DRAW, it is WIN; its DRAW child R_k cannot witness that it is WIN, so its other child is LOSS. $\square$

Proposition 5.4 (Arbitrary factorful exponent-one entry). *Let $a = J(s)$ and $k > 0$. If*

$$Q_1^e(3^k a)$$

is DRAW, then after finitely many outcome-compatible local steps either a finite provenance token has been produced, or

$$Q_2^e(a)$$

is DRAW.

Proof. Apply Lemma 5.3. If it produces a finite token, stop. If it produces the exponent-two DRAW state $Q_2^e(3^{k-1} a)$, apply Corollary 5.2. This either produces a finite provenance token or removes all remaining factors of three at the same exponent two. $\square$

6 Factor-free exponent-two base entry

It remains to analyze the base state produced by Proposition 5.4. Let $x > 0$, put

$$a = J(x), \quad P = Q_1^e(a), \quad U = Q_2^e(a),$$

$$b = B(P) = B(U), \quad F = A(U) = Q_1^e(3a), \quad g = 1 - e.$$

Suppose U is DRAW and x is the globally least coefficient source of a DRAW. The common child b has canonical coefficient strictly below $J(x)$. Since $a = J(x)$ is factor-free, this implies that the coefficient source of b is below x . Therefore b is not DRAW. As a child of a DRAW state it also cannot be LOSS; hence

$$b \text{ is WIN,} \quad F \text{ is DRAW.}$$

The exact first-factor identity is

$$\boxed{9J(x) + 1 - 2e = 2^v J(b)}. \tag{1}$$

Consequently the signed child of F is

$$A(F) = Q_v^g(J(b)).$$

Proposition 6.1 (Base-entry attachment). *Every factor-free exponent-two DRAW at a globally minimal coefficient source has a finite continuation that either strictly lowers the retained source, provides a finite proof-token descent, or enters the typed obligation normalizer through the one-shot entry component.*

Proof. We split according to the phase at x .

First suppose $e = 1 - \alpha(x)$, the B -selecting phase. The first source boundary numerator is divisible by 4, and

$$9J(x) + 1 - 2e = 3(3J(x) + 1 - 2e) - 2(1 - 2e) \equiv 2 \pmod{4}.$$

Thus $v = 1$ in (1). Writing $u = A(x)$ and using the parity defining the B -selecting phase gives the exact endpoint

$$b = 3A(x) + 1.$$

The two children of F are

$$T = Q_1^g(J(b)), \quad C = R(T),$$

and the signed-prefix identity identifies C as the ordinary child of b selected by phase g :

$$C \in \{A(b), B(b)\}.$$

At least one of T, C is DRAW. If T is DRAW, the first alternative of the obligation $O(b, g)$ holds and the typed normalizer is initialized. If C is DRAW, the other ordinary child of the WIN state b is a genuine LOSS token. For nonexceptional b , the ordinary side diamond gives a lower boundary WIN token. For the exceptional A -child orientation, the exceptional side formula gives an adjacent DRAW frame over the actual LOSS token. In the sole exceptional B -child orientation, the estimate

$$C = B(b) \leq \frac{A(b)}{8} \leq \frac{3b + 1}{16}$$

combined with $b \leq (9x + 5)/2$ yields

$$\text{src}(C) < \frac{C}{6} \leq \frac{3b + 1}{96} \leq \frac{27x + 17}{192} < x,$$

so the retained source strictly decreases.

Now suppose $e = \alpha(x)$, the A -selecting phase. Lemma 3.2 gives

$$3J(x) + 1 - 2e = 2J(A(x)),$$

and (1) therefore has $v \geq 2$. If $v \geq 4$, then

$$16J(b) \leq 9J(x) + 1 \leq 27x + 19.$$

Since $J(b) \geq 3b + 1$, we get

$$48b + 16 \leq 27x + 19, \quad b \leq \frac{9x + 1}{16} < x.$$

Thus any DRAW child of F lies below the retained source.

If $v = 2$, the two children of F are exactly

$$Q_2^g(J(b)), \quad Q_1^g(J(b)),$$

so the obligation $O(b, g)$ holds. If $v = 3$, the children form the frame

$$Q_3^g(J(b)), \quad Q_2^g(J(b)).$$

Reducing (1) modulo 3 gives

$$J(b) \equiv 1 + g \pmod{3},$$

which is precisely the hidden-parent congruence required for the typed three/two frame. That frame is therefore normalized by the same source/token routing used for typed factor frames. $\square$

7 Typed fixed-fibre normalization

The local source, factor, and proof-token diamonds used after a typed entry form a finite routing system. Its control states include A -selecting obligations, B -selecting transfers, finite factor forks, high returns, and marked terminal rows. The important distinction is between retained data and routing data.

The retained data are:

$$(\text{outer source}, \Phi(M), \eta),$$

where M is the finite multiset of carried proof-token heights and $\eta \in \{1, 0\}$ is ordered by $1 > 0$. The value $\eta = 1$ means that the current outer source/token fibre has not yet initialized the typed normalizer; the value $\eta = 0$ means that the typed normalizer is active.

Routing data include the current source cursor, phase, factor exponent, tail exponent, and local control state. These quantities may increase numerically. They do not replace the retained source unless a cited local lemma proves a strict source comparison.

Proposition 7.1 (Typed normalizer). *For every typed entry supplied with a retained arithmetic source anchor and the finite proof-token provenance required by the local diamonds, the equal-retained data routing relation is well-founded.*

Proof sketch. The finite routing system has only the following non-strict transitions inside a fixed retained fibre: bounded canonical A -streaks, valuation-two B -transfers with an installed witness, and factor-tail recurrences that carry an existing finite token. The canonical A -streak exits after at most four side tests, since four consecutive WIN states cannot occur on one A -ray. The valuation-two recycle replaces its witness by a certified lower descendant whenever it repeats. Marked factor tails either lower their tail counter or replace an existing WIN/LOSS token by lower token(s). Hence there is no equal-rank cycle in a fixed retained fibre. Strict source or token transitions are handled outside the fibre by the lexicographic rank. $\square$

Lemma 7.2 (One-shot entry). *A transition that changes η from 1 to 0 is strict regardless of the numerical value of the newly initialized inner routing source. Inside an unchanged outer source/token fibre, η never changes from 0 back to 1. A later raw reset is allowed only after a strict earlier source decrease or a strict proof-token replacement.*

Proof. Use the lexicographic order

$$(\text{outer source}, \Phi(M), \eta, \text{inner typed rank}).$$

The component $\eta : 1 \rightarrow 0$ strictly decreases after the retained source and outer token components have been fixed. Because η is placed before all inner routing data, the new inner source may be larger than the old outer source without being counted as a source update. A change $0 \rightarrow 1$ in the same outer fibre would increase the rank, so it is not a permitted routing transition. Only a strict decrease in an earlier component may reset later components. $\square$

8 Proof of the main theorem

Assume for contradiction that a DRAW exists and choose the least coefficient source s of a DRAW. The source-zero case is handled by the explicit zero-source constant-tail normalization: a zero-source DRAW reaches an exponent-one/two boundary with either a strict source exit or a typed finite terminal token.

Let $s > 0$. Take an actual DRAW whose coefficient source is s . Long-tail recurrence reduces it to tail exponent one or two after finitely many outcome-compatible moves. If a factorful exponent-one state

$$Q_1^e(3^k J(s)), \quad k > 0,$$

is reached, Proposition 5.4 gives either finite provenance for a one-shot typed entry or the factor-free exponent-two state

$$Q_2^e(J(s)) \text{ DRAW.}$$

The latter is handled by Proposition 6.1.

Thus every marked macrostep from a DRAW configuration has one of the following forms:

- (i) the retained coefficient source strictly decreases;
- (ii) at fixed retained source, an existing finite proof token is replaced by certified lower token(s), so Lemma 4.1 strictly lowers Φ ;
- (iii) the retained source and token multiset are unchanged, and the one-shot component changes $\eta : 1 \rightarrow 0$;
- (iv) $\eta = 0$ and the path remains inside the well-founded typed fibre of Proposition 7.1.

The lexicographic rank

$$(\text{retained source}, \Phi(M), \eta, \text{typed fibre rank})$$

is therefore well-founded.

On the other hand, a DRAW state has no LOSS child and at least one DRAW child. Hence after each finite macrostep one may choose an outcome-compatible continuing DRAW unless a strict smaller-source contradiction has already occurred. Repeating this choice would produce an infinite descending path in the well-founded marked relation, impossible. Hence the conjugated game has no DRAW positions.

The conjugacy of Section 2 sends the terminal conjugated state 0 to the original odd state 1. Therefore every positive odd starting position has finite remoteness and optimal play reaches 1.

9 Verification boundary and reproducibility

The proof above is a mathematical proof. The accompanying computational files are supporting checks with a narrower role:

- finite Python regressions check the arithmetic identities, valuation rows, source inequalities, and finite outcome certificates on stated finite ranges;
- the symbolic global-routing certificate checks the declared typed control/rank assembly, but it is not an end-to-end formal proof of the theorem;
- the Lean files in the repository kernel-check selected definitions and general well-founded metatheory, not the complete concrete no-DRAW theorem.

A reproducible check should record the repository revision, run the standard Python check script, run the accompanying arithmetic regression, and read the formal coverage file before interpreting the machine outputs.

References

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